

Helical Spring Motion Analysis

Problem

M4: A helical spring with a stiffness of 25 kN/m is attached to a 100 kg mass. When the mass is pulled downward and released, calculate the periodic time of its motion. Additionally, if the maximum deflection is 50 mm, determine the velocity and acceleration of the mass when it is 30 mm away from the equilibrium position.

Solution

Periodic Time of Motion

The equation for the natural frequency of a mass-spring system is given by:

$$\omega_n = \sqrt{\frac{s}{m}}$$

where:

- $s = 25 \text{ kN/m} = 25000 \text{ N/m}$ (spring stiffness),
- $m = 100 \text{ kg}$ (mass).

Substituting the values:

$$\omega_n = \sqrt{\frac{25000}{100}} = \sqrt{250} \approx 15.826 \text{ rad/s}$$

The periodic time is given by:

$$T = \frac{2\pi}{\omega_n}$$

Substituting ω_n : **(1 mark)**

$$T = \frac{2\pi}{15.826} \approx 0.397 \text{ s}$$

Velocity Calculation

Instantaneous velocity

$$v = \omega_n \sqrt{r^2 - x^2}$$

where

- $r = 50 \text{ mm} = 0.05 \text{ m}$ is the maximum deflection.
- $x = 30 \text{ mm} = 0.03 \text{ m}$ is the displacement.

$$v = 15.826 \times \sqrt{(0.05)^2 - (0.03)^2}$$

$$v = 15.826 \times \sqrt{0.0025 - 0.0009}$$

$$v = 15.826 \times \sqrt{0.0016}$$

$$v = 15.826 \times 0.04$$

$$v \approx 0.633 \text{ m/s}$$

Thus, the Instantaneous velocity is approximately **0.633 m/s. (2 marks)**

Acceleration Calculation

The acceleration equation is:

$$a = \omega_n^2 x$$

Substituting $x = 0.03$ m:

$$a = (15.826)^2(0.03)$$

$$a = (250.38 \times 0.03)$$

$$a \approx 7.513 \text{ m/s}^2$$

Thus, the acceleration of the mass at 30 mm displacement is approximately **7.513 m/s²**. **(2 marks)**